Martin Raič 5.02

računski: pisni	2 kolokvija
teoretični: ustni	
projektna naloga	

1. Centralni limitni izrek

$$S_n \sim Bin(n, \tfrac{1}{2}) \qquad P(S_n = k) = \binom{n}{k} \cdot 2^{-n}$$

$$\frac{P(S_n = k+1)}{P(S_n = k)} = \frac{\binom{n}{k+1} 2^{-n}}{\binom{n}{k} 2^{-n}} = \frac{\cancel{n!}\,(n-k)!\,k!}{(k+1)!\,(n-k-1)!\,\cancel{n!}} = \frac{n-k}{k+1}$$

Naj bo n sod $n = 2m$

$P(S_n = k)$ je maksimalna pri $k = m$ DN

(gledamo razmerje $\frac{P(S_n = k+1)}{P(S_n = k)}$)

$d = 0, 1, \ldots$

$$\frac{P(S_{2m} = m+d)}{P(S_{2m} = m)} = \frac{P(S_{2m} = m+1)\, P(S_{2m} = m+2) \cdots P(S_{2m} = m+d)}{P(S_{2m} = m)\, P(S_{2m} = m+1) \cdots} =$$

$$= \frac{2m-m}{m+1} \cdot \frac{2m-m-1}{m+2} \cdots \frac{2m-m-d+1}{m+d} =$$

$$= \frac{m \cdot (m-1) \cdot (m-2) \cdots (m-d+1)}{(m+1) \cdot \cdots \cdots (m+d)} =$$

$$= \left(\frac{1}{1+\frac{1}{m}}\right)\left(\frac{1-\frac{1}{m}}{1+\frac{2}{m}}\right) \cdots \left(\frac{1-\frac{d-1}{m}}{1+\frac{d}{m}}\right) \approx$$

$$\approx e^{-\frac{1}{m} - \frac{2}{m} - \cdots - \frac{d-1}{m} - \frac{1}{m} - \frac{2}{m} \cdots - \frac{d}{m}} = e^{-\frac{d(d-1)}{2} - \frac{d(d+1)}{2}}$$

$$\ln(1+x) \approx x \qquad 1 + x \approx e^x \qquad = e^{-\frac{d^2}{m}}$$

To velja tudi za $d = -1, -2, \ldots$ zaradi sodosti

$|d| \ll m$: $P(S_{2m} = m+d) \approx P(S_{2m} = m) \cdot e^{-\frac{d^2}{m}}$

Izkaže se: $$\lim_{m \to \infty} \sum_{d=-\infty}^{\infty} \left| \frac{P(S_{2m} = m+d)}{P(S_{2m} = m)} - e^{-\frac{d^2}{m}} \right| = 0$$

Izkaže se: $\lim_{m\to\infty} \sum_{d=-\infty}^{\infty} \left| \frac{P(S_{2m}=m+d)}{P(S_{2m}=m)} - e^{-\frac{d^2}{m}} \right| = 0$

seštevek verjetnosti

$$\sum_{d=-\infty}^{\infty} \frac{P(S_{2m}=m+d)}{P(S_{2m}=m)} = \frac{1}{P(S_{2m}=m)} \approx \sum_{d=-\infty}^{\infty} e^{-\frac{d^2}{m}}$$

$$\approx \int_{-\infty}^{\infty} e^{-\frac{x^2}{m}}\,dx = \sqrt{m}\int_{-\infty}^{\infty} e^{-t^2}\,dt = \sqrt{m\pi}$$

$x = t\sqrt{m}$

$dx = \sqrt{m}\,dt$

$$\Rightarrow P(S_{2m}=m) \approx \frac{1}{\sqrt{m\pi}}$$

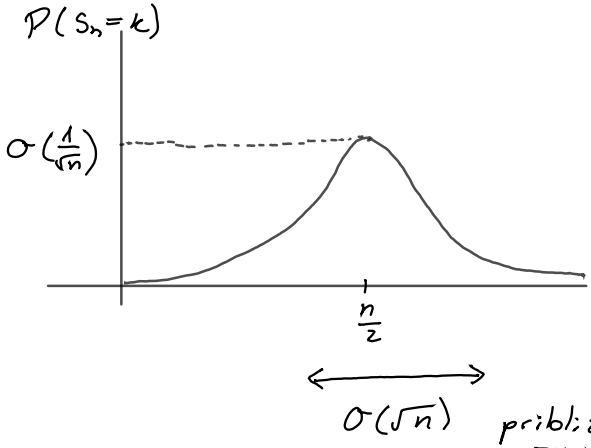
$$P(S_{2m}=m+d) \approx \frac{1}{\sqrt{m\pi}} e^{-\frac{d^2}{m}}$$

$d = k - m = k - \frac{n}{2}$:

$$\boxed{P(S_n=k) \approx \frac{1}{\sqrt{2\pi n}} e^{-\frac{2(k-\frac{n}{2})^2}{n}}}$$

De Moivreova lokalna formula

Izkaže se, da to velja tudi za lihe n



približna širina zvona

$$P(a \le S_n \le b) = \sum_{k=a}^{b} P(S_n=k) \approx \sqrt{\frac{2}{n\pi}} \sum_{k=a}^{b} e^{-\frac{2(k-\frac{n}{2})^2}{n}} \approx$$

$$\approx \sqrt{\frac{2}{n\pi}} \sum_{k=a}^{b} \int_{k-\frac{1}{2}}^{k+\frac{1}{2}} e^{-\frac{2(x-\frac{n}{2})^2}{n}} dx = \sqrt{\frac{2}{n\pi}} \int_{a-\frac{1}{2}}^{b+\frac{1}{2}} e^{-\frac{2(x-\frac{n}{2})^2}{n}} dx$$

$a, b \in \mathbb{Z} + \frac{1}{2} \quad a \le b$

$$P(a \le S_n \le b) = P(a < S_n < b) = P(a+\tfrac{1}{2} \le S_n \le b-\tfrac{1}{2}) =$$

$$= \sqrt{\frac{2}{n\pi}} \int_a^b e^{-\frac{2(x-\frac{n}{2})^2}{n}}$$

To velja tudi za splošna $a, b \in \mathbb{R}$ $a \le b$

Dobljeni približek za $P(S_n=k)$ je gostota porazdelitve $N(\frac{n}{2}, \frac{n}{4})$ v točki k.

pričakovana vrednost $\quad \sigma^2$ varianca

Podobno velja tudi za intervalske verjetnosti

Opazimo: $S \sim Bin(n,p)$ $E(S)=np$ $var(S)=np(1-p)$

Torej za $S \sim Bin(n, \frac{1}{2})$ velja $E(S)=\frac{n}{p}$ $var(S)=\frac{n}{4}$, torej enako kot za normalno porazdelitev

Binomska porazdelitev ko gre n proti neskončno je taka normalna porazdelitev, da ima isto pričakovano vrednost in varianco

Binomska porazdelitev $Bin(n,p)$
nastane kot porazdelitev vsote indikatorjev
$I_1 + \dots + I_n$ kjer so $I_1, \dots, I_n$ neodvisne
med sabo in $I_i \sim Be(p) = \begin{pmatrix} 0 & p \\ 1-p & 0 \end{pmatrix}$

Vsota veliko neodvisnih slučajnih spremenljivk z
Bernullijevo porazdelitvijo $Be(\frac{1}{2})$ je torej porazdeljena
približno normalno z ustrezno pričakovano
vrednostjo in varianco.

Primer na powerpointu
$X_1, \dots, X_n$ neodvisne. $\sim \begin{pmatrix} -1 & 0 & 9 \\ 0,5 & 0,45 & 0,05 \end{pmatrix}$

z vrjetnostjo
5% zadenemo
10 € (9 € profita)

$X_1, X_2, \dots, X_n$ neodvisne enakoporazdeljene
$E(X_1) = \mu_1 \quad E(X_1^2) < \infty \quad var(X_1) = \sigma_1^2 \quad \sigma_1 > 0$
Ce definiramo $S = X_1 + \dots + X_n$ se izkaže
$$P(a \leq S_n \leq b) \approx \frac{1}{\sigma_1 \sqrt{2\pi n}} \int_a^b e^{-\frac{(x - n\mu_1)^2}{2}} dx$$

$X_1, X_2, \dots, X_n$ neodvisne enakoporazdeljene

$E(X_1) = \mu_1 \quad E(X_1^2) < \infty \quad var(X_1) = \sigma_1^2 \quad \sigma_1 > 0$

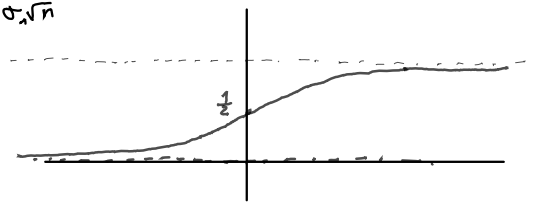
Če definiramo $S = X_1 + \dots + X_n$ se izkaže

$$P(a \leq S_n \leq b) \approx \frac{1}{\sigma_1 \sqrt{2\pi n}} \int_a^b e^{-\frac{(x-n\mu_1)^2}{2n\sigma_1^2}} dx$$

Standardizacija: $\mathbb{E} \to 0 \quad var \to 1$

$$t = \frac{x - n\mu_1}{\sigma_1 \sqrt{n}} \qquad dt = \frac{dx}{\sigma_1 \sqrt{n}}$$

$$\frac{1}{\sqrt{2\pi}} \int_{\frac{a-n\mu_1}{\sigma_1\sqrt{n}}}^{\frac{b-n\mu_1}{\sigma_1\sqrt{n}}} e^{-\frac{t^2}{2}} dt = \Phi\left(\frac{b-n\mu_1}{\sigma_1\sqrt{n}}\right) - \Phi\left(\frac{a-n\mu_1}{\sigma_1\sqrt{n}}\right)$$



$$W_n := \frac{S_n - n\mu_1}{\sigma_1 \sqrt{n}} \qquad \mathbb{E}(W_n) = 0 \quad var(W_n) = 1$$

~ približno normalno porazdeljena

$$\Rightarrow P(a < S_n \leq b) = P\left(\frac{a-n\mu_1}{\sigma_1\sqrt{n}} \leq W_n \leq \frac{b-n\mu_1}{\sigma_1\sqrt{n}}\right) \approx$$

$$P(a \leq W_n \leq b) \approx \Phi(b) - \Phi(a) = P(a \leq Z \leq b)$$

$$Z \sim N(0,1)$$

Dogovor: $Z \sim N(0,1)$

Povsod kjer je $\leq$ je lahko tudi $<$

Izrek 1.1 (centralni limitni izrek; Paul Pierre Lévy)

Naj bo $X_1, X_2, \ldots$ NEP, $S_1, S_2, \ldots$ vsote X_i, $W_1, W_2, \ldots$

$$W_n = \frac{S_n - n\mu_1}{\sigma_1 \sqrt{n}}.$$

$$\lim_{n\to\infty} P(W_n \leq w) = \Phi(w)$$

Jarl Voldemar Lindeberg:

$$\lim_{n\to\infty} \sup_{w\in\mathbb{R}} \left| P(W_n < w) - \Phi(w) \right| = \lim_{n\to\infty} \left| P(S_n \leq x) - \Phi\left(\frac{x - n\mu_1}{\sigma_1\sqrt{n}}\right) \right| = 0$$

Dokaz: opustimo (Poglej v zapiskih, ne učit na pamet)

velja za vse dokaze, ki jih opuščamo

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distribution

$$W_n \xrightarrow[n\to\infty]{d} N(0,1)$$

Izrek 1.2

Če so $W_1, W_2, \dots$ kot prej, za vsako zvezno funkcijo $h: \mathbb{R} \longrightarrow \mathbb{R}$, ki ima kvadratično omejeno rast ($\exists M \in \mathbb{R}. \forall x \in \mathbb{R}. |h(x)| \leq M(1+x^2)$) velja:

$$\lim_{n \to \infty} \mathbb{E}(h(W_n)) = \mathbb{E}(h(Z))$$

Dokaz opuščamo

$Y_1, \dots, Y_n$ neodvisne z vsoto S

$Z_1, \dots, Z_n$ neodvisne z vsoto T

zaenkrat ze n: trebada so normalne

Za $k = 1, \dots, n$ privzamemo

- $\mathbb{E}(|Y_k|^3), \mathbb{E}(|Z_k|^3) < \infty$
- $\mathbb{E}(Y_k) = \mathbb{E}(Z_k) = 0$
- $var(Y_k) = var(Z_k) = \sigma_k^2 \qquad \sigma_k \geq 0$

Tako je tudi $\mathbb{E}(S) = \mathbb{E}(T) = 0$ in

$$var(S) = Var(T) = \sigma^2 := \sum_{k=1}^{n} \sigma_k^2; \quad \sigma > 0$$

Trditev 1.3

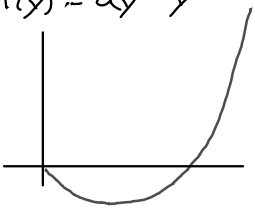
Za poljubno slučajno spremenljivko Y z vrednostmi na $[0,\infty)$ in poljuben $p \geq 1$ velja $(E(Y))^p \leq E(Y^p)$

$$\left(p=2,\ Y=|X| \quad var(X) = E(X^2) - E(X)^2 \geq 0\right)$$

Dokaz: BŠZS: $p>1$, $P(Y>0)>0$, $E(Y^p)<\infty$

Podobna neenakost $E(Y) \leq a E(Y^p)$; $a>0$

$f(y) := ay^p - y$



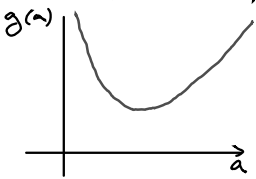
$f'(y) = apy^{p-1} - 1 = 0$

$\Rightarrow y = (ap)^{-\frac{1}{p-1}}$ minimum

$f(y) \geq f\left((ap)^{-\frac{1}{p-1}}\right)$

$ay^p - y \geq -\frac{p-1}{p}(ap)^{-\frac{1}{p-1}}$ (DN)

$$\Rightarrow E(Y) \leq aE(Y^p) + \frac{p-1}{p}(ap)^{-\frac{1}{p-1}} =: g(a)$$



$g'(a) = E(Y^p) - ap^{-\frac{p}{p-1}}$ (DN)

$ap = E(Y^p))^{-\frac{p-1}{p}}$

$$\Rightarrow E(Y) \leq g\left(\tfrac{1}{p}E(Y^p)\right)^{-\frac{p-1}{p}} \underset{\text{DN}}{=} E(Y^p)^{\frac{1}{p}}$$

QED ■

$Y_1, \dots, Y_n$ neodv. z vsoto S $\quad E(Y_k) = E(Z_k) = 0$

$Z_1, \dots, Z_n$ neodv z vsoto T $\quad var(Y_k) = var(Z_k) = \sigma_k^2$

$h: \mathbb{R} \longrightarrow \mathbb{R}$

$$E(h(S)) - E(h(T))$$

Privzemimo, da je $h \in C^3(\mathbb{R})$ in $M_3 := \sup_{x \in \mathbb{R}} |h'''(x)| < \infty$

$$\left(\begin{array}{l} h(x) = h(0) + h'(0) + \frac{1}{2} h''(0)\, x^2 + \frac{1}{6} h'''(\xi) x^3 \\ \Rightarrow h \in \mathcal{O}(x^3) \end{array} \right)$$

$$T = V_0 \longrightarrow V_1 \longrightarrow V_2 \longrightarrow \dots \longrightarrow V_n = S$$

$$V_k = Y_1 + Y_2 + \dots + Y_k + Z_{k+1} + \dots Z_n$$

Privzemimo da so $Y_1, \dots Y_n$ in $Z_1, \dots Z_n$ vsi med seboj neodvisni.

$$E(h(S)) - E(h(T)) = \sum_{k=1}^{n} E(h(V_k)) - E(h(V_{k-1}))$$

$$U_k := Y_1 + \dots + Y_{k-1} + Z_{k+1} + \dots + Z_n$$

$U_k + Y_k = V_k \qquad U_k + Z_k = V_{k-1} \qquad U_k, V_k$ in Z_k so neodvisne

Pozor! Tu nastopata U in V

$$h(V_k) = h(U_k) + h'(U_k) Y_k + \frac{1}{2} h''(U_k) Y_k^2 + R_k; \quad |R_k| \leq \frac{1}{6} M_3 |Y_k|^3$$

$$h(V_{k-1}) = h(U_k) + h'(U_k) Z_k + \frac{1}{2} h'(U_k) Z_k^2 + \tilde{R}_k$$

T is supremum

$$|\tilde{R}_k| \leq \frac{1}{6} M_3 |Z_k|^3$$

$E(h'(U_k))E(Y_k) = 0$

$$E(h(V_k)) = E(h(U_k)) + E(h'(U_k) Y_k) + \dots + E(R_k)$$

$$E(h(V_{k-1})) = E(h(U_k)) + E(h'(U_k) Z_k) + \dots + E(\tilde{R}_k)$$

$E(h'(U_k))E(Z_k) = 0$

$$E(\tilde{R}_k) \leq \frac{1}{6} M_3 E(|Z_k|^3)$$

$$E(R_k) \leq \frac{1}{6} M_3 E(|Y_k|^3)$$

$$E(h(V_k)) - E(h(V_{k-1})) \leq \frac{1}{6} M_3 \left(E(|Y_k|^3) + E(|Z_k|^3) \right) \qquad / : \sum_{k=1}^{n}$$

$$E(h(S)) - E(h(T)) \leq \frac{1}{6} M_3 \sum_{k=1}^{n} \left(E(|Y_k|^3) + E(|Z_k|^3) \right)$$